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- Problema: saber la símsa de un tanque de gas cilíndrico, con tapas planas

Datos:

$$r = 2$$

$$L = 3\text{m}$$

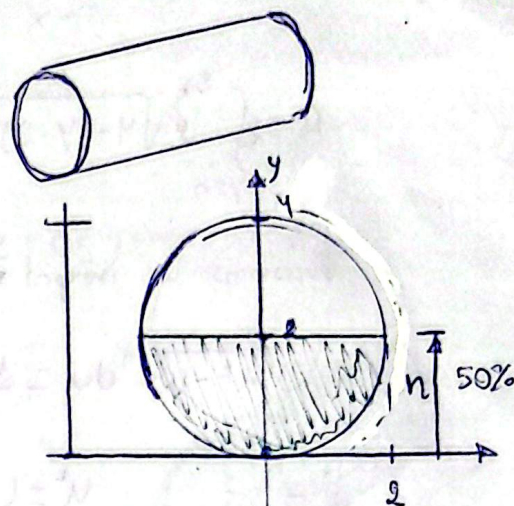
$$C = (0, 2)$$

Ecuación:

$$x^2 + (y - 2)^2 = (2)^2$$

Volumen:

$$V_T = \pi \cdot R^2 \cdot L = \pi \cdot (2\text{m})^2 \cdot 3\text{m} = 37.70\text{m}^3$$



Integral doble:

- Área de región

$$D = \{(x, y) : x^2 + (y - 2)^2 \leq 4 ; -2 \leq x \leq 2, 0 \leq y \leq 2\}$$

- Despejamos x:

$$x^2 = 4 - (y - 2)^2$$

$$x = \pm \sqrt{4 - (y - 2)^2} \Rightarrow \text{Por cada valor de } y, \text{ existen 2 puntos con la misma coordenada } y.$$

$$x \in [-\sqrt{4 - (y - 2)^2}, \sqrt{4 - (y - 2)^2}]$$

$$0 \leq y \leq 2$$

Área:

$$A = \iint_A 1 \, dA = \int_{y=0}^2 \int_{x=-\sqrt{4-(y-2)^2}}^{\sqrt{4-(y-2)^2}} 1 \, dx \, dy$$

$$\int_{x=-\sqrt{4-(y-2)^2}}^{\sqrt{4-(y-2)^2}} 1 dx = \sqrt{4-(y-2)^2} - (-\sqrt{4-(y-2)^2}) = 2\sqrt{4-(y-2)^2}$$

$$A = 2 \int_{y=0}^2 \sqrt{4-(y-2)^2} dy = 2 \left[\right]$$

Buscamos una integral tabulada parecida:

$$\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \arcsen\left(\frac{u}{a}\right) + C$$

$$a^2 = 2^2$$

$$a = 2$$

$$u^2 = (y-2)^2$$

$$u = (y-2)$$

$$F(u) = \frac{u}{2} \sqrt{4-u^2} + \frac{4}{2} \arcsen\left(\frac{u}{2}\right)$$

Cambiamos los límites $0 \leq y \leq 2$ a los de u .

$$u = (0-2) = -2$$

$$u = (2-2) = 0$$

} nuevos límites

$$F(0) = 0 + 2 \arcsen(0) = 0$$

$$F(-2) = \frac{-2}{2} \sqrt{4-(-2)^2} + 2 \arcsen\left(\frac{-2}{2}\right) = 0 + (-\pi) = -\pi$$

$$A = 2 \cdot [F(0) - F(-2)] =$$

$$A = 2 \cdot [0 - (-\pi)] = 2\pi$$

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Volumen Total =

$$V = \int \int \int 1 \, dx \, dy \, dz$$

- Volumen de Region:

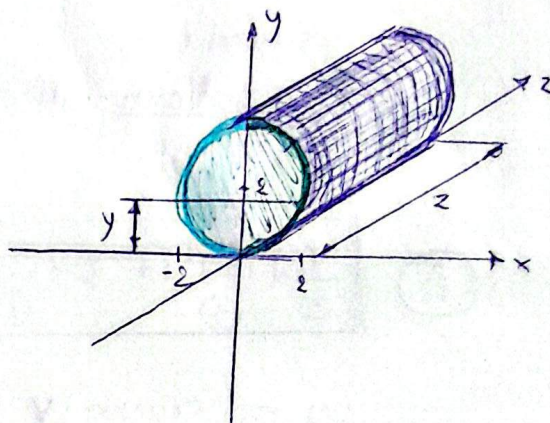
$$V = \{ (x, y, z) : x^2 + (y-2)^2 \leq 4 ; -2 \leq x \leq 2 ; 0 \leq y \leq 4 ; 0 \leq z \leq 3 \}$$

$$x = \pm \sqrt{4 - (y-2)^2}$$

$$\begin{cases} x = -\sqrt{4 - (y-2)^2} \\ x = \sqrt{4 - (y-2)^2} \end{cases}$$

$$0 \leq y \leq 4$$

$$0 \leq z \leq 3$$



$$V = \iiint_V 1 \, dV = \int_{z=0}^3 \int_{y=0}^4 \int_{x=-\sqrt{4-(y-2)^2}}^{\sqrt{4-(y-2)^2}} 1 \, dx \, dy \, dz$$

$$V_{\text{Total}} = \int_{y=0}^4 \int_{z=0}^3 \left[2\sqrt{4-(y-2)^2} \right] dz \, dy$$

$$V_{\text{Total}} = \int_{y=0}^4 \left. 2\sqrt{4-(y-2)^2} z \right|_0^3 dy = \int_{y=0}^4 6\sqrt{4-(y-2)^2} dy$$

Integral reducida: cambio de coordenadas

$$\int \sqrt{4-(y-2)^2} dy = \frac{u}{2} \sqrt{4-u^2} + \frac{4}{2} \arcsin\left(\frac{u}{2}\right) \Big|_{y=0}^{y=4}$$

Cambio de limites:

$$u = y-2 \quad \begin{cases} u = 0-2 = -2 \\ u = 4-2 = 2 \end{cases}$$

$$u_{\text{max}} = 4-2 = 2$$

Definimos una variable: v

$$v = y-2 \quad \rightarrow \quad u \in [-2; 2] \quad \rightarrow \quad 0 \leq y \leq 4$$

$$v \in [-2; 2]$$

$$V = 6 \int_{u=-2}^2 \sqrt{4-u^2} du$$

calculamos en los extremos: $u = -2$; $u = y_0 - 2 = V$

$$F(-2) = \frac{-2}{2} \sqrt{4 - (-2)^2} + \frac{4}{2} \arcsen\left(\frac{-2}{2}\right) = -\pi$$

$$F(-2) = -\pi$$

$$F(V) =$$

$$V_{dato} = 6 \cdot [F(V) - F(-2)] = 6 [F(V) - (-\pi)] = 6F(V) + 6\pi$$

Despejamos $F(V)$

$$F(V) = \frac{V_{dato}}{6} - \pi$$

$$\textcircled{A} \quad \boxed{\frac{V}{2} \sqrt{4 - V^2} + 2 \arcsen\left(\frac{V}{2}\right) = \frac{V_{dato}}{6} - \pi} \quad \text{Donde } V \in [-2; 2]$$

Una vez calculado V se recupera y

$$y = y_0 - 2 \rightarrow y_0 = V + 2$$

* Método Numérico para resolver el valor de V

- ya que V es una función trascendental, no se puede despejar V en forma convencional

- tenemos que eliminar la raíz y el arcsen.

Cambio de variables:

$$\textcircled{1} \quad V = 2 \cdot \text{sent}$$

$$\frac{2 \cdot \text{sent}}{2} \sqrt{4 - (2 \cdot \text{sent})^2} = \text{sent} \cdot [2 \cos t] = 2 \text{sent} \cdot \cos t = \text{sen}(2t)$$

$$\textcircled{2} \quad 2 \arcsen\left(\frac{2 \cdot \text{sent}}{2}\right) = 2 \cdot \underbrace{\arcsen(\text{sent})}_{\substack{\text{Inversa y forma} \\ \text{normal de coseno}}} = 2 \cdot t$$

$$A := \text{Sen}(2t) + 2t = \frac{V_{dato}}{6} - \pi$$

- Límites de t : $0 \leq y \leq 4 \rightarrow v[-2/2]$

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$$t = \arcsen\left(\frac{y}{2}\right) \left\{ \begin{array}{l} t = \arcsen\left(-\frac{2}{2}\right) \rightarrow t = -\frac{\pi}{2} \\ t = \arcsen\left(\frac{2}{2}\right) \rightarrow t = \frac{\pi}{2} \end{array} \right\} t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

constante

$$C = \frac{V_{dato}}{6} - \pi$$

Ecuación:

$$\sen(2t) + 2t = C \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Definimos la función:

$$\phi(t) = \sen(2t) + 2t - C = 0$$

- Ecuación generica: con $x_1 \leq x \leq x_2$; $0 \leq y \leq y_2$; $0 \leq y \leq z_2$; R
 $L_{max}(2R)$

$$V_{dato} = \int_{y_1}^{y_2} (z_2 - 2) \sqrt{R^2 - (y - R)^2} dy$$

Cambio de coordenadas

$$u = y - R \quad du = dy$$

Límites máximos

$$\left. \begin{array}{l} u = 0 - R = -R \\ u = 2R - R = R \end{array} \right\} u \in [-R, R]$$

Límite y a altura H

$$u = H - R \quad H \in [0, 2R]$$

$$V_{dato} = 2 \cdot z_2 \int_{u=-R}^{u=H-R} \sqrt{R^2 - u^2} du$$

- Definimos una variable V

$$V = H - R$$

$$V \in [-R; R]$$

$$a^2 = R^2$$

$$a = R$$

- Evaluamos la integral entre los extremos ; $-R \leq u \leq V$

$$F(-R) = -\frac{R}{2} \sqrt{R^2 - (-R)^2} + \frac{R^2}{2} \arcsen\left(-\frac{R}{R}\right) = F_{-u_0} = \frac{R^2}{2} \arcsen\left(-\frac{R}{R}\right)$$

$$F(V) = \frac{V}{2} \sqrt{R^2 - V^2} + \frac{R^2}{2} \arcsen\left(\frac{V}{R}\right)$$

$$V_{\text{dato}} = 2 \cdot Z_e [F(V) - F(-R)]$$

Despejamos $F(V)$

$$F(V) - F(-R) = \frac{V_{\text{dato}}}{2 \cdot Z_e}$$

$$F(V) = \frac{V_{\text{dato}}}{2 \cdot Z_e} + F(-R)$$

$$\frac{V}{2} \sqrt{R^2 - V^2} + \frac{R^2}{2} \arcsen\left(\frac{V}{R}\right) = \frac{V_{\text{dato}}}{2 \cdot Z_e} + F(-R)$$

- Cambio de variable t . $V = R \cdot \text{sen } t$

$$\sqrt{R^2 - V^2} = \sqrt{R^2 - (R \cdot \text{sen } t)^2} = \sqrt{R^2 (1 - \text{sen}^2 t)} = R \cdot \text{cost}$$

$$\textcircled{1} \left(\frac{R \cdot \text{sen } t}{2}\right) (R \cdot \text{cost}) = \frac{R^2}{2} (\text{sen } t \cdot \text{cost}) = \frac{R^2}{2} \frac{1}{2} \text{sen}(2t)$$

identidad trigonométrica

$$\text{sen } t \cdot \text{cost} = \frac{1}{2} \text{sen}(2t)$$

$$\textcircled{2} \frac{R^2}{2} \arcsen\left(\frac{R \cdot \text{sen } t}{R}\right) = \frac{R^2}{2} t$$

$$A = \frac{R^2}{4} \text{sen}(2t) + \frac{R^2}{2} \overset{\text{arcsen}}{t} = \frac{V_{\text{dato}}}{2 \cdot Z_e} + F(-R)$$

constante:

$$C = \frac{V_{Dato}}{2Z_2} + F(-R) = \frac{V_{Dato}}{2Z_2} + \frac{R^2}{2} \arcsen(-1) = \frac{V_{Dato}}{2Z_2} + \frac{R^2}{2} \left(-\frac{\pi}{2}\right)$$

Ecuación

$$\frac{R^2}{4} \sin(2t) + \frac{R^2}{2} t = C$$

①

②

- Derivados Por el Metodo de Newton-Rapson

$$\textcircled{1} \left[\frac{R^2}{4} \sin(2t) \right]' = \frac{2 \cdot R^2}{4} \cos(2t) = \frac{R^2}{2} \cos(2t)$$

$$\textcircled{2} \left[\frac{R^2}{2} \arcsen\left(\frac{R \cdot \sin t}{2}\right) \right]'$$

$$u = \frac{R}{2} \sin t$$

$$\frac{d}{du} \left[\frac{R^2}{2} \arcsen(u) \right] \frac{d(u)}{dt} \rightarrow \text{Regla de la cadena}$$

$$\textcircled{2} \left(\frac{R^2}{2} t \right)' = \frac{R^2}{2}$$

$$\frac{R^2}{2} \frac{1}{\sqrt{1-u^2}} \cdot \frac{d}{dt} \left[\frac{R \sin t}{2} \right]$$

$$\frac{R^2}{2} \frac{1}{\sqrt{1-u^2}} \cdot \frac{R \cos t}{2} = \frac{R^3}{4} \frac{\cos t}{\sqrt{1-\left(\frac{R \sin t}{2}\right)^2}}$$

Definimos la función

$$g(t) = \frac{R^2}{2} \cos(2t) + \frac{R^3}{4} \frac{\cos t}{\sqrt{1-\left(\frac{R \sin t}{2}\right)^2}} = 0$$

$$g(t) = \frac{R^2}{4} \sin(2t) + \frac{R^2}{2} t - C = 0$$

$$g(t) = \frac{R^2}{4} \sin(2t) + \frac{R^2}{2} t - \frac{V_{Dato}}{2Z_2} + \frac{R^2}{2} \frac{\pi}{2} = \frac{R^2}{2} \left[\frac{\sin(2t)}{2} + t + \frac{\pi}{2} \right] - \frac{V_{Dato}}{2Z_2}$$

$$0 \leq y \leq \pi$$

$$g'(t) = \frac{R^2}{2} \cos(2t) + \frac{R^2}{2} = 0$$

$$\frac{R^2}{2} (\cos(2t) + 1) = 0$$

Sabemos que

$$V = H - R$$

— Cuando hallamos t con un método numérico, reemplazamos:

$$V = R \cdot \text{sent}$$

y luego:

$$H = V + R$$

$$\Rightarrow H = R \cdot \text{sent} + R$$